INCH-POUND
MIL-I-14802A
31 January 1990
SUPERSEDING
MIL-I-14802(MU)
14 February 1969 and
Amendment 2
15 July 1986

MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M3A2, REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the remanufactured M3A2 Cartridge Actuated Initiator Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 Quality Assurance Terms and Definitions
MIL-STD-413 Visual Inspection Guide for Elastomeric O-rings

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:47Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CAGE Code 19200 and 19203)

8595822	Initiator, Cartridge Actuated, Delay, Metal Parts Assembly
8796541	Box, Fiberboard (Inner) Packing, Ammunition, for Initiator, Cartridge Actuated, Delay, M3A2 or M5A2
8796542	Box, Packing, Ammunition, for Initiator, Cartridge Actuated, Delay, M3A2 or M5A2
9216934	Box, Fiberboard (Outer) Alternative Packing, Ammunition, for Initiator, Cartridge Actuated, M3A2 or M5A2

U.S. AIR FORCE

TPO 00-932-5031	Initiator, Cartridge Actuated, M3A2

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (3720), Indian Head, MD 20640-5000.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured initiator metal parts assembly shall contain a new initiator spring, safety pin seal and O-rings.

3.2 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component, regardless of configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (see 6.3.2).

3.3 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 5500 to 6500 pounds per square inch (psi) for 15 seconds, minimum (min), without deformation or mechanical failure (see 6.3.1).

3.4 Firing pin release. The firing pin of the assembled initiator, less safety pin, cartridge retainer and chamber shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin ¾ inch, maximum (max).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

3.5 Initiator pin security. The initiator pin of the assembled initiator, less cartrige retainer and chamber, shall not withdraw beyond the safety pin when subjected to an axial load of 200 to 220 pounds. Application of the load shall not cause permanent deformation of the safety pin. The safety pin groove shall be 0.118 +0.010, −0.000 inch. Each safety pin groove shall undergo 100% inspection.

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

a. When the remanufacturing and loading of assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.7 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using the procedures in 4.5.5.

3.8 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspection. Inspection shall be in accordance with MIL-A-2550.

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:47Z
Check the source to verify that this is the currentversion before use.

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 68 to 77 °F
b. Altitude: Sea level
c humidity: 50 percent relative, max
d. Atmospheric pressure: 14.7 psia

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.3.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies, max, provided the homogeneity requirements of inspection lots have been retained.

4.3.2 Examination. One-hundred percent examination of reprocessed parts shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability.

4.3.2.1 Examination of reclaimed parts. Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.3.2.2 Housing, Firing Pin, Drawing 8596169 covering a detail of Drawing 8595822.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Pitch diameter of small external thread, min	Gage
102	Major diameter of small external thread, min	Gage
103	Pitch diameter of large external thread, min	Gage
104	Major diameter of large external thread, min	Gage
105	Thread damaged	Visual
106	Metal defective	Visual
107	Protective finish inadequate or defective	Visual
108	Surface finish of cavity improper	Visual
109	Burr or sharp corner	Visual
110	Foreign matter present	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:47Z
Check the source to verify that this is the current version before use.

Characteristic	Defect	Method of Inspection
Minor		
201	Pitch diameter of small external thread, max	Gage
202	Pitch diameter of large external thread, max	Gage
203	Surface finish improper, except as otherwise classified	Visual

4.3.2.3 Pin, Firing, Drawing 8596168 covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish improper	Visual
104	Burr or sharp edge	Visual
Minor		
None defined		

4.3.2.4 Pin, Initiator, Drawing 8595403 covering a detail of Drawing 8595822.

Critical		
1	Diameter of groove	Gage
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish on body and body groove improper	Visual
104	Burr	Visual
105	Foreign matter present	Visual
Minor		
201	Surface finish improper, except as otherwise classified	Visual

4.3.2.5 Cap, Drawing 8595404 covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Pitch diameterof large internal thread, max	Gage
102	Minor diameter of large internal thread, max	Gage
103	Pitch diameter of small internal thread, max	Gage
104	Minor diameter of small internal thread, max	Gage
105	Safety pin groove missing	Visual
106	Thread damaged	Visual
107	Protective finish inadequate or defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

Characteristic	Defect	Method of Inspection
108	Burr	Visual
109	Foreign matter present	Visual
110	Metal defective	Visual/X-ray
Minor		
201	Pitch diameter of large internal thread, min	Gage
202	Pitch diameter of small internal thread, min	Gage

4.3.2.6 Retainer, Cartridge, Drawing 8593013 covering a detail of Drawing 8595822.

Critical		
1	Metering slots in base missing	Visual
Major		
101	Pitch diameter of internal thread, max	Gage
102	Minor diameter of internal thread, max	Gage
103	Elongated slot missing	Visual
104	Thread damaged	Visual
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Burr	Visual
108	Foreign matter present	Visual
Minor		
201	Pitch diameter of internal thread, min	Gage

4.3.2.7 Chamber, Drawing 8593014 covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Pitch diameter of large external thread, min	Gage
102	Major diameter of large external thread, min	Gage
103	Pitch diameter of small external thread, min	Gage
104	Major diameter of small external thread, min	Gage
105	Thread damaged	Visual
106	Metal defective	Visual
107	Protective finish inadequate or defective	Visual
108	Surface finish, end of gas port, improper	Visual
109	Burr	Visual
110	Foreign matter present	Visual
Minor		
201	Pitch diameter of large external thread, max	Gage
202	Pitch diameter of small external thread, max	Gage
203	Surface finish improper, except as otherwise classified	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:47Z
Check the source to verify that this is the current version before use.

4.3.2.8 Assembly, Drawing 8595822, less Drawings 8595405, 8593013, 8593014, and MIL-C-5501/11-F4.

Characteristic	Defect	Method of Inspection
Critical		
1	Protrusion of firing pin	Gage
2	Steel ball missing	Gage
3	O-ring missing	Visual
Major		
None defined		
Minor		
201	O-ring not lubricated	Visual

4.3.2.9 Assembly, Drawing 8595822.

Critical		
None defined		
Major		
101	Initiator pin not free to rotate	Manual
102	Safety pin loose or deformed	Manual/Visual
103	Any part missing	Visual
Minor		
201	Protective cap missing	Visual
202	Protective finish inadequate or defective	Visual

4.3.2.10 Box, Fiberboard (Inner) Packing Ammunition Drawing 8796541.

Critical		
None defined		
Major		
None defined		
Minor		
201	Gap between body and cover, max	Manual/Visual
202	Contents loose	Manual
203	Metal ends loose or distorted	Manual
204	Cut or damaged through all impregnated layers	Visual

4.3.2.11 Box, Packing Ammunition, Drawing 8796542.

Critical		
None defined		
Major		
None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

Characteristic	Defect	Method of Inspection
Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken or improperly assembled	Manual/Visual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.3.2.12 Box, Fiberboard (Outer) Alternative, Packing Drawing 9216934.

Critical

None defined

Major

None defined

Minor		
201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.3.2.13 Pin, Safety, Drawing 8595405 covering a detail of Drawing 8595822.

Critical

None defined

Major		
101	Diameter of wire	Gage
102	Profile incorrect	Visual
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp edges	Visual
106	Foreign matter present	Visual

4.3.2.14 Spring, Initiator, Drawing 8595406 covering a detail of Drawing 8595822.

Critical

None defined

Major		
101	Load at specified heights	Gage
102	Height, solid, max	Gage
103	Ends incorrect	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp edges	Visual
106	Foreign matter present	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

Characteristic	Defect	Method of Inspection
Minor		
201	Diameter outside, closed, max	Gage
202	Diameter inside, min	Gage
203	Free length	Gage

4.3.2.15 Seal, Drawing 10524102 covering a detail of Drawing 8595822.

Critical		
None defined		
Major		
101	Diameter outside, min	Gage
102	Width, min	Gage
103	Thickness, min	Gage
104	Cut, cracked or split	Visual
105	Flash	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter outside, max	Gage
202	Width, max	Gage
203	Thickness, max	Gage

4.4. Inspection testing. The following tests shall be performed in accordance with 4.5.

4.4.1. Hydrostatic pressure. Each chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.3 shall be cause for rejection of the subassembly.

4.4.2. Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.3. Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit in the inspection sample to comply with 3.7 shall be cause for rejection of the assembly.

4.5 Test methods and procedures.

4.5.1 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.2 Hydrostatic pressure. Prior to the application of protective finish, the cap and chamber shall be mounted in the device, using a fixture for properly sealing all openings and subjected to the internal hydrostatic pressure to determine compliance with 3.3.

4.5.3 Firing pin release. The initiator pin of each subassembly shall be subjected to the firing pin release test. Failure of any subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

4.5.4 Initiator pin security. The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any subassembly to comply with 3.5 shall be cause for rejection of the subassembly.

4.5.5 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degrees rotation of the assembly for four full passes.

4.6 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be levels A or B as specified by the contracting activity:

Level A — In accordance with Drawings 8796541 and 8796542
Level B — In accordance with Drawings 8796541 and 9216934

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-932-5031.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

6.2 Acquistion requirements. Acquisition documents must specify the following;

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)
c. The responsible inspection activity (see 4.1)
d. Level of marking and packing (see 5.1).

6.3 Definitions.

6.3.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3.2 Remanufactured exterior components. Remanufactured exterior components are: cap, firing pin housing, and chamber.

6.4 Subject term (keyword) listing

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:47Z
Check the source to verify that this is the current version before use.

Life-saving systems, aircrew

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy-OS
Air Force-99

Review activity:
Air Force-70

Preparing activity:
Navy-OS
(Project 1377-0960)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-14802A	900131

3. DOCUMENT TITLE

INITIATOR, CARTRIDGE ACTUATED, M3A2, REMANUFACTURED METAL PARTS ASSEMBLY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Commanding Officer Naval Ordnance Station	(301) 743-4250	364-4250
c. ADDRESS (Include Zip Code) (Code 3730) Indian Head, MD 20640-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 20250918164472
Check the source to verify that this is the latest version.